The Hyperspace Recursive Intelligence Hypothesis: can Bayesian model comparison distinguish real cosmogenesis?

A Bayesian and physical research programme

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Abstract

How could a physical hypothesis of intelligent cosmogenesis become distinguishable from its rivals? This paper separates the Hyperspace Recursive Intelligence Hypothesis into real daughter-universe creation, multi-generation lineages and a stronger closed-cycle submodel. It combines channel-specific physics, Bayesian model comparison, observer weighting and branching processes. The observer-count limit is a conditional mixture identity, not a universal probability ceiling; a Poisson lineage has positive survival probability only under its stated stationarity and independence assumptions above unit mean offspring. The proposed discriminator is maker-selection beyond habitability, tested against life-only, natural-reproductive and technology-only alternatives with independently frozen factors. Initial order remains a candidate observation, but set containment does not determine its likelihood ratio. Five study specifications and seven failure conditions define what could reject each claim. No cosmological posterior, demonstrated creation channel or empirical validation is reported. The programme's AI-alignment work and laws under test are separate: they neither require this extension nor establish it. Physical feasibility, observer measures and external specialist review remain open.

1 Introduction

The programme begins with a practical question: if control of advanced AI is lost against our will, what could preserve useful and aligned behaviour? Human control should be retained for as long as possible. This paper addresses a separate, conditional extension: whether an intelligence could instantiate a physically real daughter cosmology, and what evidence could distinguish that account from its rivals. A cosmologist assessing a candidate transition channel, or a statistician comparing selection models, can use the definitions and failure conditions below without accepting the hypothesis. No ARC law or result establishes the extension.

The paper connects cosmology and quantum gravity, Bayesian model comparison, observer selection, branching processes and thermodynamics. The long-form HRIH edition preserves the broader hypothesis and its dated record; this companion isolates the physical and statistical questions. The introduction states the three nested propositions; Sections 6 and 8 specify the channel and selection requirements, and Section B gives the proposed studies. Their outcomes have not been obtained here.

1.1 Contributions

The contribution is a reviewable research architecture: nested real-cosmogenesis, lineage and closed-cycle models; channel-specific feasibility conditions; a conditional branching threshold; maker-selection beyond

Other fields; working paper. Conditional cosmological extension of the ARC Theory; none of Laws I to V is a premise or tested here. Version 4.2, 2 November 2026; doi:10.17605/OSF.IO/4PGVC. Theory registration and definitions; dated manuscript record. Michael Darius Eastwood, BA (Hons), without an institutional affiliation or doctorate: author and record; Paper C. Working draft; open review in Section 9. CC BY 4.0.

habitability; explicit observer weighting; and comparison with natural and simulation families. It supplies no empirical posterior. Section 2 identifies earlier work on these components; priority for their precise conjunction remains unresolved pending the systematic review listed in Section 9.

1.2 The origin question that remains after early-universe cosmology

Modern cosmology describes the evolution of an early hot, dense universe with extraordinary success. Inflationary and quantum-cosmological proposals address important features of the early state and, in some models, remove an ordinary temporal boundary. They do not all answer the same question. A model may describe how a universe can emerge without specifying why this set of laws, constants and boundary data is realised. HRIH targets that selection question.

The scientifically meaningful form of the question is not simply "why is there something rather than nothing?" No current physical theory eliminates all primitives. The tractable question is:

> Why do the laws and initial conditions of our universe permit a history containing observers, technological intelligence and, perhaps, access to cosmogenic processes; and is the joint permission of those outcomes better predicted by deliberate maker-selection than by non-deliberate cosmology or observer selection alone?

1.3 The nested hypothesis

HRIH is not one indivisible proposition. It is a hierarchy (Figure 1):

- **R1. Real intelligent cosmogenesis.** At least one physically admissible process allows an advanced agent in a parent history to instantiate a physically real daughter cosmology.
- **R2. Recursive lineage.** Real intelligent cosmogenesis occurs in more than one generation and can be represented as a branching lineage.
- **R3. Closed-cycle HRIH.** The complete cosmogenic history satisfies a global fixed-point or periodic-boundary condition, so the Big Bang is an internal temporal boundary of a self-consistent whole rather than an unexplained first event in an external linear chain.

Evidence for R1 does not entail R2; evidence for R2 does not entail R3. A confirmed baby-universe channel would therefore be scientifically important without proving the closed-loop ontology.

1.4 What "hyperspace" means in this paper

HRIH is retained as the historical name of the hypothesis. In scientific propositions, *hyperspace* does not assert extra dimensions, a particular metric, a superluminal channel or a known quantum-gravitational mechanism. It denotes an as-yet-unspecified cosmogenic channel. A physical submodel becomes scoreable only when it supplies a state space, dynamics or action, boundary conditions, parameter domain and an observational map. Until then, the neutral term *unknown cosmogenic channel* is used.

1.5 The strongest claim and its honest limit

The strongest HRIH claim is that a globally self-consistent cosmogenic history could explain why the Big Bang need not be an uncaused first event in a linear sequence. This is a genuine explanatory structure. It does not answer why the space of possible laws, the global consistency rule or the admissible solution exists at all. The paper therefore claims a candidate closure of a causal regress, not an explanation without primitives.

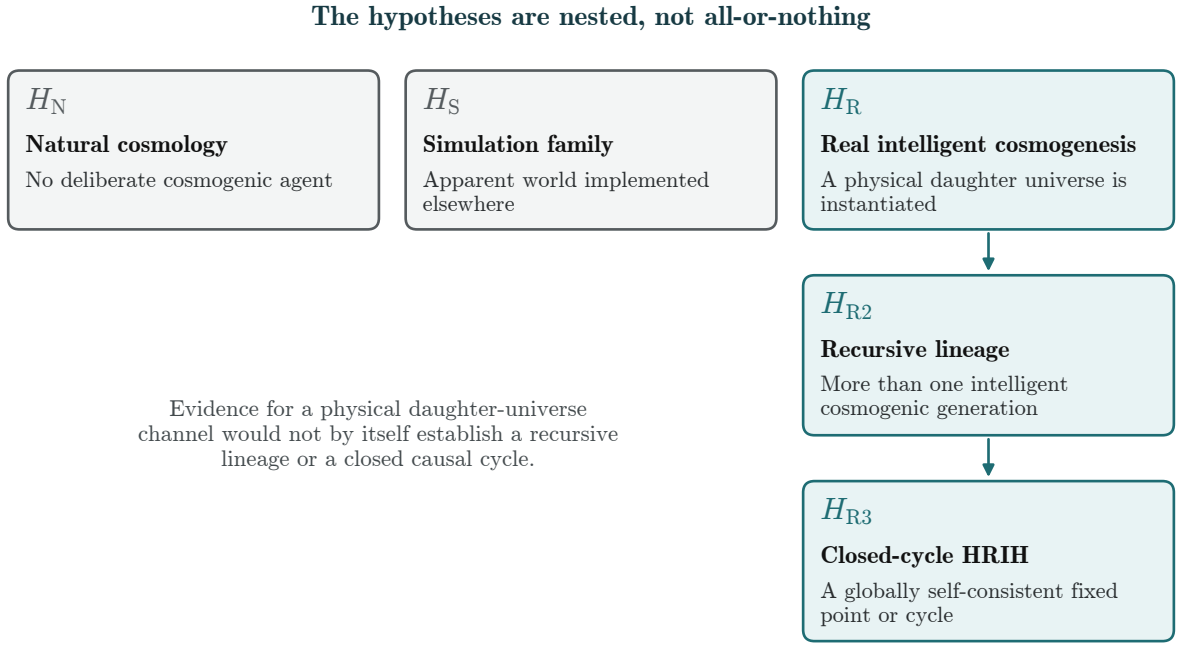


Figure 1. Nested model space. The closed-cycle model is the strongest and most fragile HRIH submodel, not a premise smuggled into the weaker physical-cosmogenesis hypothesis. H_{R2} and H_{R3} label the submodels of H_{R} that propositions R2 and R3 state.

Withdrawal of an invalid inference. The existence of intelligence in our universe does not imply a prior creator, an infinite lineage or a closed loop. Recursion alone does not remove the need for a base case, boundary condition, fixed point or global consistency rule. Closed-cycle HRIH is a postulate to be tested for existence, stability, physical admissibility and predictive value.

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2 Related work

2.1 Intelligent and reproductive cosmology

HRIH belongs to a small but real lineage of cosmological proposals involving universe reproduction or intelligent participation. Smolin (Smolin 1992) proposed cosmological natural selection through black holes. Crane's meduso-anthropic principle (Crane 1994), Harrison's intelligent-life selection proposal (Harrison 1995), Gardner's *Biocosm* (Gardner 2003), and Vidal's inquiry into whether natural or artificial selection could hold on a cosmic scale (Vidal 2014) discuss intelligence and universe creation in different forms. These are antecedents, not embarrassments. HRIH's defensible novelty must be stated as a precise remainder after they are credited.

2.2 Quantum cosmology and no-boundary proposals

Hartle-Hawking no-boundary cosmology (Hartle and Hawking 1983), Vilenkin's tunnelling proposal (Vilenkin 1982), and past-eternal or null-boundary models such as Aguirre-Gratton (Aguirre and Gratton 2003) show that the absence of an ordinary initial temporal boundary is not unique to HRIH, while the positive averaged-expansion condition in the theorem of Borde, Guth and Vilenkin implies past geodesic incompleteness (Borde, Guth, and Vilenkin 2003). HRIH's distinctive ambition is not merely to remove a boundary mathematically, but to model selected initial conditions through intelligent cosmogenesis and, in R3, global closure.

2.3 Anthropic reasoning and observer measures

Carter (Carter 1974), Weinberg (Weinberg 1987) and later multiverse work developed observer-selection arguments; Srednicki and Hartle (Srednicki and Hartle 2010) made explicit that a theory in a large universe requires a xerographic distribution. HRIH therefore cannot treat "we observe a life-permitting universe" as direct evidence of deliberate creation. Its maker-selection claim must predict something beyond observer permission.

2.4 Simulation

Bostrom's argument (Bostrom 2003), Chalmers's philosophy of virtual reality (Chalmers 2022), Kipping's Bayesian model (Kipping 2020) and physical critiques (Vazza 2025) provide a mature comparison literature. HRIH should use that literature as a methodological benchmark, not as an opponent to defeat rhetorically.

The six-part contribution is stated once in Section 1.1. The comparisons above identify adjacent work, not a completed priority search.

3 Transition, lineage and closure, stated as mathematics

3.1 Universe histories and cosmogenic transitions

Let X_g denote the physically relevant state of a universe history in generation g. It can include laws and constants θ_g, initial quantum state ψ_g, low-energy effective fields, and any inherited data admitted by a particular physical channel. Let A_g denote properties of a cosmogenic agent or civilisation arising in X_g. A candidate transition kernel is

$$X_{g+1} \sim \mathcal{T}_\chi\big(\cdot \mid X_g, A_g, \phi_g\big)\,, \tag{1}$$

where χ identifies the physical channel and ϕ_g its parameters. Equation (1) is not evidence that such a transition exists. It is the minimum mathematical object needed to state the hypothesis without converting a metaphor into a conclusion.

The transition decomposes into at least six contingent events:

(a) intelligence arises;

(b) it reaches relevant cosmogenic capability;

(c) the proposed physical channel is real and accessible;

(d) the agent chooses real cosmogenesis rather than alternatives;

(e) the resulting daughter is viable;

(f) the creator system remains sufficiently stable to complete the act.

Each event requires an explicit probability or physical model. None may be assigned probability one by definition.

3.2 A closed-cycle submodel

A closed-cycle history can be represented as a fixed point

$$X^\star = C\left(X^\star; \phi\right), \tag{2}$$

or as a periodic orbit

$$X_{g+L} = X_g \tag{3}$$

for some $L \geq 1$, with generation $g + L$ identified with generation g, so that the history closes on itself rather than repeating without end. Here C includes both ordinary physical evolution and the cosmogenic transition. This formulation does not require a signal travelling backwards through local laboratory time. It treats the complete history as a boundary-value solution whose regions are jointly constrained.

For Eq. (2) or (3) to become physics, a submodel must establish:

1. **existence**: at least one admissible solution;
2. **measure**: non-zero amplitude or probability under the theory;
3. **stability**: robustness to perturbations;
4. **causal consistency**: no contradiction with the channel's causal structure;
5. **observational content**: a likelihood distinguishable from rival cosmologies.

Self-consistency inside an assumed loop proves none of these by itself.

3.3 Occurrence is not constructibility

The occurrence of our universe establishes that a universe history exists. It does not establish that agents inside a universe can reproduce the relevant boundary conditions. Natural events frequently depend on regimes inaccessible to internal engineering. The constructibility hypothesis is therefore channel-specific:

$$H_{\text{construct}}(\chi): \quad P(\text{successful daughter cosmology} \mid A, \chi) > 0. \tag{4}$$

A single counterexample to one channel does not exclude all channels; conversely, merely naming an unknown channel is not a rescue. Each scored channel must carry an explicit physical model and a complexity penalty (Simpson, Rue, Riebler, Martins, and Sø rbye 2017).

3.4 Ability is not preference

Even if real cosmogenesis is feasible, a capable agent may prefer simulation, virtual worlds, observation, reversible experiments, or no creation at all. Let

$$p_W = P(\text{agent chooses physically real cosmogenesis} \mid \text{capable, informed}). \tag{5}$$

HRIH does not set $p_W = 1$. Preference is a behavioural and normative uncertainty that must be modelled, bounded or measured. This term sharply distinguishes real intelligent cosmogenesis from a simulation argument.

4 Bayesian model space and conditional observer counting

4.1 Top-level families

For a specified inference, first name the focal experienced world and the level of explanation. The following families distinguish its immediate implementation or origin:

$$H_{\mathrm{N}} : \text{directly physical history, not deliberately seeded,} \tag{6}$$
$$H_{\mathrm{S}} : \text{experienced world implemented by external computation,} \tag{7}$$
$$H_{\mathrm{R}} : \text{directly physical daughter, deliberately instantiated.} \tag{8}$$

These are categories for the chosen focal level, not an exhaustive ontology. Simulation and real cosmogenesis can coexist: a simulation may run inside a deliberately created physical universe. Such ancestry belongs in an explicit hybrid submodel. If the focal definitions do not make alternatives disjoint, compare joint submodels indexed by implementation and origin rather than adding overlapping family probabilities. The sums below require mutually exclusive model assignments, normalised weights and a declared treatment of omitted possibilities.

Each family must contain explicit submodels. H_{N} may include a single-history model, inflationary or quantum-cosmological boundary proposals, multiverse selection (Bousso and Polchinski 2000; Garriga and Vilenkin 2008), and natural cosmological reproduction (Smolin 1992). H_{S} must state the simulated reference class, computational substrate and fidelity. H_{R} includes one-generation cosmogenesis, recursive lineages and the closed-cycle submodel.

An unspecified "unknown mechanism" has no scoreable likelihood. Flexibility is handled through declared, proper priors and normalised likelihoods, not by an automatic penalty assigned in prose.

4.2 Posterior model probabilities

For data D, model parameters ϑ_j and prior model weights π_j,

$$P(H_j \mid D) = \frac{\pi_j \int p(D \mid \vartheta_j, H_j) p(\vartheta_j \mid H_j)\, \mathrm{d}\vartheta_j}{\sum_k \pi_k \int p(D \mid \vartheta_k, H_k) p(\vartheta_k \mid H_k)\, \mathrm{d}\vartheta_k}. \tag{9}$$

The integrals are marginal likelihoods. They reward predictive concentration and penalise unused flexibility. No headline probability is scientifically meaningful unless the model space, priors, evidence clusters and observer measure are presented beside it.

4.3 Comparison with the simulation argument

Bostrom's simulation argument (Bostrom 2003) is a conditional trilemma about civilisation survival, willingness to run ancestor simulations and observer counts; it is not a direct measurement of whether we are simulated. Bayesian treatments such as Kipping's (Kipping 2020) obtain near-balanced odds only inside specified partitions and priors. Physical analyses such as Vazza's (Vazza 2025) impose resource constraints on simulations of this universe. HRIH is different: it requires a physical cosmogenesis channel and a maker-selection likelihood. A number from the simulation literature cannot be transferred to HRIH, and the two hypotheses cannot be ranked responsibly by counting how many independent authors have assigned them percentages.

The proper comparison is Eq. (9), with each family carrying its own physical constraints, reference class and complexity cost. At present, HRIH has no registered empirical posterior. This paper deliberately supplies a model, not a favourable number.

4.4 The observer-count limit depends on model weights

Bostrom's indifference step assigns the fraction of observers with the relevant experiences to the corresponding credence, conditional on no distinguishing information (Bostrom 2003). Kipping explicitly model-averages over the possibility that simulations are not produced and states the dependence on model priors (Kipping 2020). The following elementary identity makes that dependence visible; it is not a new universal probability ceiling.

Proposition 1 (Conditional count mixture)**.** *Let G be a creator-type family. Assume one uncreated reality and N created realities inside G, equal relevant observer weight per reality, and no created observers outside G. If $w = P(G \mid D)$ and the within-family observer fraction conditional on D remains $N/(N+1)$, then*

$$P(\text{created} \mid D) = w\frac{N}{N+1} \quad \longrightarrow \quad w \quad (N \longrightarrow \infty). \tag{10}$$

The law of total probability gives this result; Appendix A records the derivation and its limits. Without model-discriminating evidence, $w = \pi$ recovers the prior-weighted illustration in Table 1. Evidence that changes the family weight changes the limit. Unequal observer weights change the fraction.

Created realities N	$\pi = 1/2$	$\pi = 1/3$	$\pi = 1/6$
1	0.2500	0.1667	0.0833
10	0.4545	0.3030	0.1515
1,000	0.4995	0.3330	0.1665
1,000,000	0.4999995	0.3333330	0.1666665
Limit	1/2	1/3	1/6

Table 1. Conditional arithmetic, not measured probabilities. Each column is $\pi N/(N+1)$ under Proposition 1, with no model-discriminating evidence. The columns change the assumed mass, not an astronomical observation.

For two mutually exclusive creating families in an equally weighted three-family illustration, their combined mass is $2/3$; it is only this stipulated case that approaches $2/3$. Refining a family into submodels while preserving its total prior mass changes no probability. Assigning equal weights anew after drawing more boxes changes the prior and must be reported as such. The same requirements apply to any proposed HRIH probability.

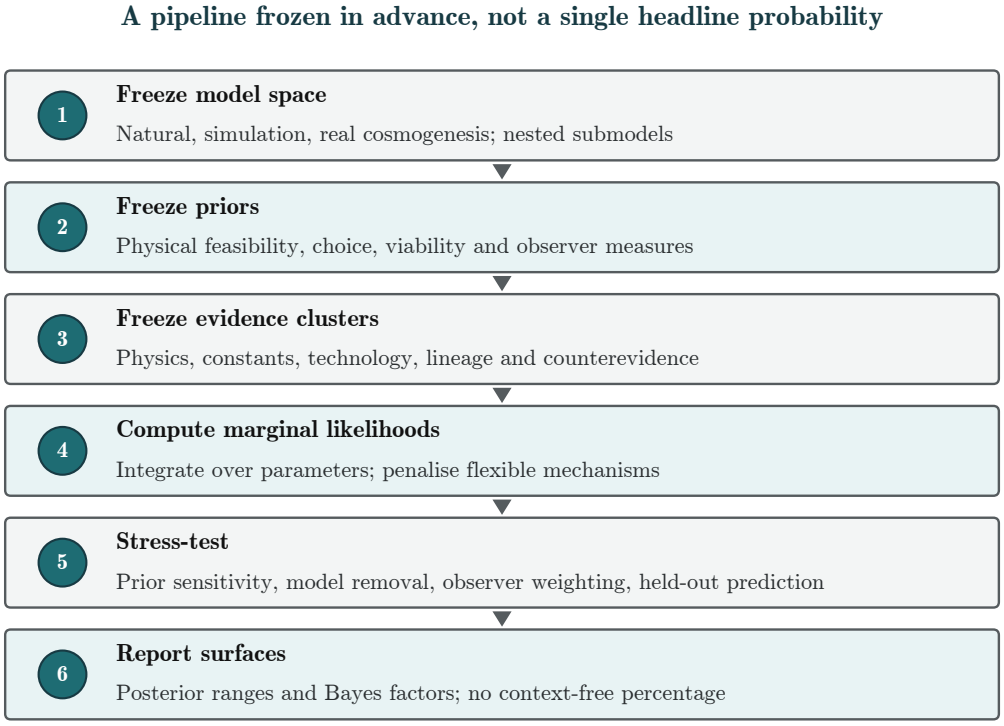


Figure 2. The required Bayesian workflow. The output is a sensitivity surface over declared models and assumptions, not an author's intuitive probability converted into decimals.

4.5 Prior discipline

The workflow in Figure 2 is supplemented by prior-predictive checks and an explicit dependence model for reused evidence. Prior families and observer measures must be defensible before outcomes are inspected. All predeclared model-removal, cluster-removal and observer-weighting sensitivities are reported.

5 How observers are weighted can change the posterior

5.1 Why physical theory is not enough

In a large or branching cosmology, a physical theory can generate many observers or observer-moments compatible with some data. Prediction then requires a rule for locating "us" among them (Hartle and Srednicki 2007). Let O_j be the observer set under H_j and $\xi_j(o)$ a xerographic distribution (Srednicki and Hartle 2010). Then

$$p(D \mid H_j, \xi_j) = \sum_{o \in O_j} \xi_j(o)\, p(D \mid o, H_j). \tag{11}$$

Different ξ_j can materially change the posterior. This is not a technical nuisance. It is one of the central uncertainties in multiverse, simulation and recursive-lineage inference.

5.2 Reference classes to freeze

The model-space study must decide, in the protocol it registers, whether it weights:

- observers or observer-moments;

- civilisations or individuals;
- biological, artificial or both kinds of intelligence;
- root and descendant universes equally or by physical measure;
- exact duplicates separately;
- simulated and physically embodied observers in a shared or separate class.

No choice is neutral. HRIH should therefore report a multiverse of observer measures and identify which conclusions survive all reasonable variants.

5.3 Indexical evidence

Observing a life-permitting universe is conditioned by the existence of an observer. Its evidential strength still depends on the normalised rival likelihoods and the observer measure; selection alone does not establish that the evidence is weak or neutral. A proposed discriminator should identify what each rival predicts after applying the same declared observer condition. This motivates maker-selection in Section 8.

6 Physical feasibility is scored one channel at a time

6.1 Why a physical channel is indispensable

A cosmogenic hypothesis acquires scientific content only through a concrete channel. The fact that general relativity, semiclassical gravity or speculative quantum gravity contains unusual solutions does not establish that an internal agent can reach them. A channel is scoreable only if it specifies:

1. the relevant fields and state variables;
2. the action, Hamiltonian or transition rule;
3. boundary and initial conditions;
4. parameter ranges and approximation regime;
5. resource requirements accessible to the agent;
6. the mapping from parent controls to daughter observables;
7. failure modes and a likelihood function.

6.2 False-vacuum bubble and tunnelling channels

Farhi and Guth (Farhi and Guth 1987) showed a classical obstruction to creating an inflating universe in the laboratory under their assumptions. Later work by Farhi, Guth and Guven (Farhi, Guth, and Guven 1990), and by Fischler, Morgan and Polchinski (Fischler, Morgan, and Polchinski 1990a; Fischler, Morgan, and Polchinski 1990b), examined tunnelling and Hamiltonian treatments of false-vacuum bubbles. These papers are serious antecedents, not confirmations of HRIH. They expose precisely the problems the hypothesis must confront: singularity theorems, topology change, pseudomanifold constructions, boundary conditions, negative-energy or mass issues, and the validity of semiclassical approximations.

For a semiclassical tunnelling submodel the minimum quantitative object is a rate of the form

$$\frac{\Gamma}{V} = A \exp\left(-\frac{B}{\hbar}\right), \tag{12}$$

where B is an appropriate Euclidean action difference and A a prefactor. HRIH-B, the proposed physical-channel study, must choose one channel, state A, B, boundary conditions and validity range, and permit the

resulting rate to become negligibly small. If the calculation cannot be completed without an uncontrolled construction, that submodel remains unscored.

6.3 Black-hole and bounce channels

Cosmological natural selection (Smolin 1992) and meduso-anthropic proposals (Crane 1994) consider black holes as potential reproductive sites. These ideas motivate, but do not establish, parameter inheritance or agent control. A maker-selection model needs an explicit map

$$\theta_{g+1} \sim K_{\chi}(\theta_{g+1} \mid \theta_g, a_g) \tag{13}$$

showing which parent variables can influence which daughter variables, with what noise. If no information or control can traverse the relevant boundary, maker-selection is physically inoperative even if a daughter region exists.

6.4 Chronology and global closure

The closed-cycle submodel is more demanding than a lineage. It may be formulated as a global boundary condition without local backwards signalling, but it still requires a physical theory admitting the complete solution. Closed timelike curve models and quantum consistency analyses (Deutsch 1991) provide mathematical precedents, while Hawking's chronology-protection analysis (Hawking 1992) supplies a serious counterpressure through quantum backreaction. HRIH does not claim that chronology violation is established or required. The closed-cycle model must be separately penalised and separately killable.

6.5 Physical resource accounting

Simulation and cosmogenesis have different resource profiles. Vazza's astrophysical constraints (Vazza 2025) apply to computing a sufficiently detailed simulation on a physical substrate; real cosmogenesis may instead attempt to seed a state that evolves under its own dynamics. That distinction can reduce one cost while introducing much harder gravitational constraints. HRIH must not use the resource cost of simulation as evidence for real cosmogenesis. It must calculate the cost and feasibility of the chosen real channel.

Useful constraints include entropy and information bounds, thermodynamic costs, available free energy, stability of the parent vacuum, concentration of energy, and causal access to the relevant region. Bekenstein, holographic and computational bounds (Bekenstein 1981; Susskind 1995; Lloyd 2002) are boundary conditions, not evidence that a universe can be engineered. Table 2 sets the candidate channels side by side.

6.6 Duration-dependent costs and autonomous instantiation

Vazza's Planck-resolution Earth calculation gives 2.55×10^{59} erg at the cosmic microwave background temperature. He compares this with roughly 5.6×10^{59} erg for the rest-mass energy of Palomar 2, with mass about 3.3×10^5 solar masses, and estimates a similar expenditure per simulation timestep; approximately 10^6 such timesteps reach the Milky Way comparison (Vazza 2025). These are his model-specific resource estimates, not a bound on every possible simulation or every possible host physics.

A conditional cost model makes the distinction precise. Suppose a host pays $c > 0$ per simulated timestep of fixed duration Δt, and a hypothetical autonomous daughter requires only a one-off parent cost

Candidate channel	What would make it scoreable	Principal present obstacle
False-vacuum bubble	Explicit action, prefactor, accessible initial state, daughter mapping	Singularities, topology change, semiclassical control, extreme suppression
Black-hole / bounce	Quantum-gravity transition and inheritance kernel	No accepted mechanism or controllable parameter transfer
Closed-cycle boundary	Admissible global solution, non-zero measure, stability and observable consequences	Chronology protection, measure, underdetermination
Simulation	Computational substrate, fidelity, reference class and resource budget	Energy/information cost; does not produce a physically real daughter
Unknown channel	Full state space, dynamics and observational map	Otherwise unlimited flexibility and no likelihood

Table 2. Physical channel discipline. An unmodelled possibility remains an open placeholder, not a favourable submodel.

E_{seed}. In a continuous interpolation,

$$\begin{aligned} E_{\text{sim}}(T) &= E_0 + cT/\Delta t, & \frac{\partial E_{\text{sim}}}{\partial T} &= c/\Delta t > 0, \\ E_{\text{inst}}(T) &= E_{\text{seed}}, & \frac{\partial E_{\text{inst}}}{\partial T} &= 0. \end{aligned} \tag{14}$$

The second line is an assumed parent-cost boundary, not a demonstrated cosmogenesis channel. It requires no continued support, control, maintenance or duration-dependent resource demand from the daughter. Relaxing any of those assumptions changes the model. The first line also assumes fixed fidelity, positive expenditure per step and unchanged accounting; it is not a technology-independent theorem. The daughter's own energy and entropy accounting remains part of its physics.

Faizal and colleagues argue that limits on algorithmic completeness preclude a fully algorithmic account of reality and draw an anti-simulation conclusion (Faizal, Krauss, Shabir, and Marino 2025). That conclusion is their argument. Formal undecidability alone does not establish that a specified finite observation cannot be simulated. Nor does a constraint on computation establish or exclude an agent-accessible cosmogenesis channel. This paper therefore transfers neither their conclusion nor Vazza's resource calculation into evidence for HRIH.

The comparison retains two distinct burdens: simulation must specify fidelity, substrate and computational resources; instantiation must specify an admissible and accessible gravitational channel, its one-off cost and any continuing costs. The singularity and semiclassical obstacles above remain. No ordering of the families' probabilities follows from comparing an energy estimate with an unresolved channel.

7 Lineage survival in a conditional branching model

7.1 Definition

Let the expected number of viable daughter universes generated by one universe history be

$$\mu_R = p_A p_C p_U p_W p_V p_K m_R, \tag{15}$$

where:

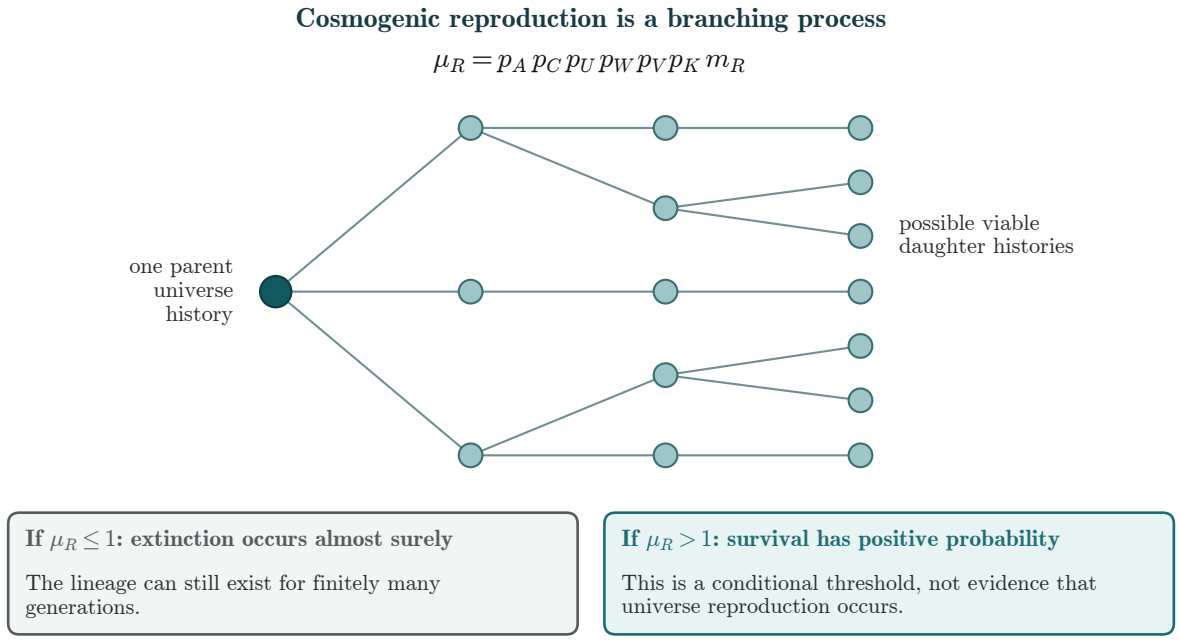


Figure 3. A hypothetical branching lineage. The displayed threshold assumes the stationary, single-type, independent Poisson model of Proposition 2; it is not a universal criterion for correlated or changing environments.

p_A	probability that relevant intelligence arises;
p_C	probability that it reaches cosmogenic capability;
p_U	probability that an accessible real-universe channel exists;
p_W	probability that capable agents choose to use it;
p_V	probability that a generated daughter is viable;
p_K	probability that the creator system remains sufficiently stable;
m_R	expected number of attempted daughters, conditional on successful parent preparation and before applying per-daughter viability.

Equation (15) is a factorised baseline, not a measurement or an identity for arbitrary joint distributions. Here m_R counts attempted daughters conditional on successful parent preparation; p_V is the per-daughter viability probability, so viability is counted once. Under conditional sequential probabilities, the parent-event product can instead be written by the chain rule; otherwise dependencies require an explicit joint expectation. Correlation between yield and viability cannot be discarded. No factor is fixed to one by definition.

7.2 Extinction and survival

Proposition 2 (Poisson lineage threshold)**.** *Start from one ancestor in a single-type Galton-Watson process. Assume independent offspring counts with one stationary Poisson distribution of finite mean μ_R for every parent and generation. The extinction probability q is the smallest solution in $[0, 1]$ of*

$$q = \exp[\mu_R(q - 1)]\,. \tag{16}$$

Thus

$$\mu_R \leq 1 \quad \Longrightarrow \quad q = 1, \tag{17}$$

and a lineage becomes extinct almost surely. If $\mu_R > 1$, survival has positive probability $1 - q$ (Figure 3).

This standard branching-process result (Harris 1963) is derived for the stated Poisson model in Appendix A. It establishes no physical reproduction process. Shared environments, multiple types and changing offspring laws require a different analysis; a marginal average above one is not then a sufficient survival condition.

7.3 What the threshold contributes

The threshold converts a vague narrative of recursive creation into a set of measurable or constrainable terms. A theory can fail because the channel is physically impossible, because capable agents rarely choose it, because daughters are non-viable, because creator civilisations are unstable, or because the expected yield is too low. The model is valuable even if the posterior mass ends up concentrated below one.

7.4 Closed cycles are not required for branching

A recursive lineage may have a finite or unknown origin. The branching programme therefore survives rejection of the closed-cycle submodel. Conversely, $\mu_R > 1$ does not supply a closed cycle: a closed cycle requires a compatible global boundary condition and a path through state space returning to the required conditions. These are distinct hypotheses and must never be reported as one confirmation.

8 Maker-selection beyond habitability is the principal discriminator

8.1 The discriminator

Let θ denote physical laws, constants and boundary parameters. Let $p_0(\theta)$ denote a baseline distribution supplied by a natural-cosmology submodel. Define:

$$L(\theta) = \text{observer or life permission}, \tag{18}$$

$$C(\theta) = \text{accessibility of cosmogenic technology}, \tag{19}$$

$$S(\theta) = \text{probability of stable technological ascent}, \tag{20}$$

$$K(\theta) = \text{expected viable daughter yield}. \tag{21}$$

A simplified life-conditioned natural model has

$$p(\theta \mid H_{\mathrm{N}}, \mathrm{obs}) \propto p_0(\theta) L(\theta), \tag{22}$$

whereas a simplified HRIH maker-selection model predicts

$$p(\theta \mid H_{\mathrm{R}}, \mathrm{obs}) \propto p_0(\theta) L(\theta) C(\theta) S(\theta) K(\theta). \tag{23}$$

Both expressions require finite, positive normalising integrals over the same declared parameter measure. With $M(\theta) = C(\theta)S(\theta)K(\theta)$ and normalisers Z_N and Z_R, their pointwise density ratio on common support is $M(\theta)Z_N/Z_R$. Extra positive factors alone therefore imply no evidential advantage at the observed parameters.

The distinctive claim is not that our universe permits life. It is that the realised parameter region is unusually favourable to technologically accessible, stable and viable cosmogenesis beyond what life-only, black-hole-production and technology-only selection models predict.

8.2 Independence requirements

The functions C, S and K must be frozen before inspecting the held-out target relationships. They may use training data to define physically meaningful proxies, but they cannot be reverse-engineered from the constants they are supposed to explain. A valid study should:

1. predefine parameter ranges and rival models;
2. select creator-enabling proxies using physical theory independent of the held-out outcomes;
3. fit on a training subset of parameter relationships;
4. predict held-out relationships or sensitivity directions;
5. compare out-of-sample likelihood against life-only, black-hole-selection and technology-only models;
6. report model and prior sensitivity.

8.3 Candidate creator-enabling factors

Potential factors include long-lived free-energy gradients; stable complex chemistry; compact objects and high-energy environments; metastable but sufficiently durable vacuum structure; laws permitting information processing and error correction; and physical access to a cosmogenic channel. These are candidates, not a confirmed list. Some are already selected by ordinary habitability, and others may be generic consequences of broad parameter ranges. HRIH earns evidence only from the residual predictive contribution after those overlaps are modelled.

8.4 Rival selection models

Maker-selection is adjacent to established and speculative predecessors:

- anthropic and observer selection (Carter 1974; Weinberg 1987);
- Smolin-style cosmological natural selection through black-hole production (Smolin 1992);
- Crane, Harrison, Gardner and Vidal-style intelligence-related cosmogenesis (Crane 1994; Harrison 1995; Gardner 2003; Vidal 2014);
- a technology-only selection model, defined here as a rival, in which laws favour increasingly capable observers.

The paper's potentially distinctive remainder is the joint inclusion of creator-system stability and a quantitative comparison of maker-selection against both life-only and non-intelligent reproductive models. Priority for that precise remainder remains a literature-search question, not a premise of the physics.

8.5 Initial order as a proposed discriminator

Penrose's estimate of the specialness of the initial condition, one part in $10^{10^{123}}$, is conditional on his phase-space account (Penrose 2004). His earlier argument asks why an observer-producing fluctuation should account for the much larger observed entropy imbalance; his proposed initial Weyl-curvature condition is a distinct answer (Penrose 1979), developed later in conformal cyclic cosmology (Penrose 2010). This paper uses neither the estimate nor that objection as a measured probability for HRIH.

Maker-selection asks a further question: does producing a long-lived technological lineage require a distribution of initial order distinguishable from that required by the observer condition alone? It is plausible that sustained free-energy gradients, cumulative selection and stable information processing matter. Landauer's principle bounds heat dissipation in logically irreversible erasure under its physical

assumptions (Landauer 1961); it does not supply a lower bound on the initial entropy of an entire maker-producing universe, or a cost for every learning operation.

Let E be a pre-specified measurable event or finite-resolution region for initial order, and define

$$\Lambda_{\mathcal{R}}(E) = \frac{P(E \mid H_{\text{mak}})}{P(E \mid H_{\text{obs}})}, \qquad P(E \mid H_{\text{obs}}) > 0. \tag{24}$$

Every maker-containing history contains observers under the chosen definition. That set inclusion, even combined with a higher minimum order for makers, does not establish $\Lambda_{\mathcal{R}}(E) > 1$ for the observed region. Two normalised distributions over histories and an observation model are needed. Appendix A gives a finite counterexample: containment can coexist with a ratio below, equal to or above one.

Defeaters retained for a future specified model.

E1 If a rival observer-selection model predicts the observed order as well as the maker model, the proposed discriminator supplies no advantage.

E2 If a maker can arise without more initial order than the comparator requires, the proposed ordering premise fails.

E3 If a dynamical initial-condition model predicts the observation, compare that model explicitly; its success may remove the proposed selection advantage. Inflation alone is not assumed to settle this question.

The likelihood ratio is presently undetermined against both fluctuation-only and inflationary observer-selection alternatives. The former edition's strict inequality is withdrawn. The physical question and the three defeaters remain; no numerical posterior or favourable sign is inferred.

9 What this paper does and does not show

9.1 Evidence clusters

The Bayesian programme should divide evidence into clusters whose dependencies can be inspected:

1. **Physical feasibility**: channel-specific theoretical and experimental constraints.
2. **Parameter distribution**: fine-tuning and held-out maker-selection predictions.
3. **Astrophysical structure**: black-hole production, vacuum stability and energy resources.
4. **Technological trajectory**: evidence bearing on p_C, p_W and p_K.
5. **Indexical evidence**: observer and reference-class information.
6. **Counterevidence**: constraints that exclude or suppress candidate channels.

Evidence reused in more than one cluster must be explicitly linked so it does not update the model twice.

9.2 Current evidence status

At present:

- no accepted physical channel has been shown to permit agent-controlled creation of a viable daughter universe;
- no observation identifies a parent universe or deliberate parameter selection;
- no registered maker-selection model has outpredicted life-only or natural reproductive rivals;
- no observer measure is uniquely justified;

- no lower-level ARC result supplies a likelihood ratio for HRIH's cosmological ontology.

No HRIH-specific likelihood has been supplied for broad ARC observations. The defensible action is to make no cosmological update from them. A neutral factor can record that convention,

$$\mathrm{BF}_{R/N}^{\text{bookkeeping}}(D_{\mathrm{ARC}}) := 1, \tag{25}$$

but it is not an estimated Bayes factor or a theorem that the unknown likelihoods are equal. A specified generative bridge, fixed before the relevant analysis, is needed to make that comparison empirical.

9.3 Open review and edition status

This is version 4.2, a working draft for author review. It retains the open work recorded in version 4.1: review from outside the programme by a statistician, a cosmologist or quantum-gravity specialist and a philosopher of probability; a systematic review of the precise contribution's priority; and the reference-code deposit described in Appendix D. None is represented as completed by this typesetting and correction pass. The paper remains a working draft while those checks are open. Appendix J records the substantive corrections and the relationship to the separate long-form edition.

9.4 Compatibility is not support

HRIH may be compatible with general relativity, quantum mechanics, evolution, information theory and diverse philosophical or religious traditions. Compatibility means that no immediate contradiction has been identified at the stated level. It does not increase the probability of HRIH. Support requires a new derivation, quantitative prediction or likelihood not already supplied by the established theory.

9.5 An evidence ledger

Every evidence item should carry the following fields:

Claimed relevance	Which HRIH submodel and parameter it bears on
Source	Primary source, dataset and immutable identifier
Status	Existing, forecast, correspondence, counterevidence or unscored
Likelihood	How the item changes $p(D \mid H_j)$, if at all
Dependence	Other evidence clusters sharing the same information
Rival prediction	What H_{N}, H_{S} and adjacent selection models predict
Kill condition	What outcome makes the item evidence against HRIH

Scientific standing depends on the specified physical and statistical tests; dates, branding and biography supply no cosmological likelihood.

10 Five predictions, each a hypothesis with what would count against it

Each of P1 to P5 is a hypothesis to be decided by the studies of Section B, stated with what would count against it; none is a forecast of a study's outcome, and this paper attaches no confidence to any, because no registered model yet exists to supply one.

10.1 Primary maker-selection prediction

P1: held-out maker-selection. A model using independently defined $C(\theta)$, $S(\theta)$ and $K(\theta)$ predicts held-out relations among physical parameters better than life-only anthropic selection, cosmological natural selection and technology-only rivals. Failure to outperform those models counts against HRIH; redefinition of the factors after failure is prohibited.

10.2 Physical-channel prediction

P2: at least one scoreable channel. At least one physically specified cosmogenesis channel retains non-negligible transition probability over an agent-accessible parameter region after singularity, topology, backreaction, resource and validity constraints are imposed. If every well-specified channel is excluded or negligibly suppressed across defensible ranges, real intelligent cosmogenesis is strongly disfavoured.

10.3 Branching prediction

P3: supercritical region. For the stationary independent Poisson lineage in Proposition 2, indefinite survival requires a non-negligible parameter region with $\mu_R > 1$. If the posterior mass for $\mu_R > 1$ becomes negligible under all defensible priors and dependencies, indefinite survival in those specified submodels is disfavoured even if isolated cosmogenesis remains possible.

10.4 Closed-cycle prediction

P4: admissible closure. The closed-cycle submodel requires at least one stable, non-zero-measure solution of Eq. (2) or (3) in a specified physical theory and at least one observational consequence not shared by the corresponding open lineage. Failure removes the closed-cycle explanation without removing R1 or R2.

10.5 Preference prediction

P5: real-creation choice cannot be assumed. HRIH's real-cosmogenesis posterior must integrate over p_W. Evidence that capable agents overwhelmingly prefer simulation, abstention or non-cosmogenic goals lowers the real-lineage probability. The paper makes no claim that ability entails preference.

11 What would count against HRIH, and what would not

11.1 Scientific kills and model-specific kills

A scientific theory must be allowed to lose probability. The following outcomes have different scopes and must not be conflated:

K1. Channel exclusion. Mature physical analysis excludes every specified agent-accessible real-cosmogenesis channel, or drives its transition rate to negligible values across the declared parameter domain. This kills the scored real-cosmogenesis submodels, not every logically imaginable mechanism.

K2. Maker-selection failure. Independently frozen maker-selection factors fail to improve held-out prediction beyond life-only, natural reproductive and technology-selection rivals, with an equivalence region ruling out a scientifically meaningful advantage. This kills the distinctive maker-selection claim.

K3. Subcritical lineage. Within the stationary independent Poisson model, the posterior probability of $\mu_R > 1$ becomes negligible under every defensible joint parameter model. This kills the corresponding indefinite-survival claim while leaving isolated creation, and lineages of finitely many generations, possible.

K4. Closure exclusion. The closed-cycle model has no admissible, stable, non-zero-measure solution, or has no observational distinction from an open lineage after complexity penalties. This kills R3.

K5. Observer-measure fragility. HRIH is favoured only under a narrow, outcome-selected xerographic distribution. This counts against a robust cosmological inference.

K6. Prior-volume fragility. Apparent support disappears under modest and defensible prior or model-space changes. This counts against a claimed posterior rather than being hidden in a supplement.

K7. Preference suppression. Empirical or theoretical evidence drives p_W close to zero. This can make real intelligent cosmogenesis rare even if the physical channel exists.

11.2 What is not a scientific kill

An earlier document containing the same idea would defeat a priority claim but would not falsify HRIH. Failure of an ARC experiment would alter only the bridge or stability factors it actually informs. Rejection of the cosmological closed loop would not refute a one-generation physical cosmogenesis channel. These distinctions are essential to prevent both overclaiming and artificial survival by moving between claim types.

11.3 No branch may turn every outcome into success

A fired registered kill condition kills the claim as stated. It may motivate a new hypothesis, but the new hypothesis must receive a new identifier, date and registration. A null result is not support; a confidence interval containing the null is not equivalence; and a statistically detectable but scientifically negligible effect does not establish the proposed mechanism.

12 Limitations

12.1 Physics is incomplete

Quantum gravity is unresolved. Semiclassical calculations may fail in the regimes most relevant to cosmogenesis. HRIH cannot bridge those gaps by naming hyperspace or by invoking future intelligence. Unknown physics can keep a possibility open; it cannot make it probable.

12.2 Priors and measures dominate

Model priors, parameter priors and observer measures can dominate cosmological posteriors; for anthropic predictions, Starkman and Trotta argue that the answer depends on how probabilities are assigned to possible universes (Starkman and Trotta 2006). The paper's remedy is sensitivity and registration, not a claim that one choice is uniquely correct.

12.3 One observed universe

We have one observed universe and strong selection effects. Many proposed discriminators may remain underpowered or underdetermined. Maker-selection must therefore focus on structured, held-out

predictions rather than visual closeness to familiar constants.

12.4 Technological forecasting

Cosmogenic capability depends on existential risk, coordination, energy, time and a feasible physical channel. Near-term AI progress constrains none of these by itself.

12.5 Normative uncertainty

p_W may be small. Advanced agents may reject real cosmogenesis for ethical or practical reasons. HRIH cannot assume that creation is inevitable merely because it is possible.

12.6 The closed loop may remain metaphysical

If R3 produces no observational distinction from an open lineage or a natural boundary model, it cannot become an empirically preferred cosmological theory even if it remains logically coherent. The paper permits that outcome.

13 Conclusion

HRIH is not established cosmology. It does not presently show that intelligence created our universe, that the Big Bang was engineered, or that a beginningless intelligent loop is physically real. Its central intuition nevertheless survives in a scientifically stronger form.

A physically real daughter-universe channel is a legitimate question at the frontier of gravitation and quantum cosmology. If intelligence can access such a channel, cosmogenic histories can be represented as branching processes. If creator-enabling properties affect the distribution of daughter constants, maker-selection can in principle be compared with life-only and natural reproductive selection. If a specified global theory admits a stable fixed point or cycle, the Big Bang can become an internal boundary of a self-consistent history rather than an unexplained first event in a linear chain.

The work ahead is explicit: choose one physical channel; freeze a fair model space; specify observer measures; compute branching survival; define maker-selection factors independently; and permit every central claim to lose. The most defensible HRIH is therefore not a declaration that the origin problem has been solved. It is a Bayesian and physical research programme whose strongest submodel supplies a candidate solution and whose lower rungs can be tested separately.

Code and data

This edition is a document and mathematics review. It reports no new study execution or data analysis. Appendix D describes the separately proposed reference implementation; its deposit remains open.

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A Conditional derivations and counterexample

These arguments check the mathematical statements made in the paper. They are not a simulation or an analysis of cosmological data.

A.1 Observer-count mixture

For the event C of being created by family G's mechanism, total probability gives

$$P(C \mid D) = P(C \mid G, D)P(G \mid D) + P(C \mid G^c, D)P(G^c \mid D). \tag{A.1}$$

Under Proposition 1, the second term is zero and the first conditional probability is $N/(N+1)$. This proves Equation (10). If $\pi = P(G)$ and Z_G, Z_{G^c} are positive finite marginal likelihoods,

$$P(G \mid D) = \frac{\pi Z_G}{\pi Z_G + (1-\pi)Z_{G^c}}. \tag{A.2}$$

It need not equal π. For unequal observer masses W_C and W_0, replace $N/(N+1)$ by $W_C/(W_C+W_0)$. A refinement into disjoint submodels preserves a family's probability when their masses sum to its original mass; merely relabelling the same alternatives does not create evidence.

A.2 Poisson extinction

Let $F(z) = \exp[\mu_R(z-1)]$ be the Poisson probability-generating function. Independence and stationarity give extinction by generation n as $F^{\circ n}(0)$, an increasing bounded sequence. Its limit is a fixed point and cannot exceed any non-negative fixed point, hence is the smallest one. For $0 < \mu_R \leq 1$, $F'(z) = \mu_R F(z) \leq 1$ on $[0,1]$ and $F(z) - z > 0$ for $z < 1$, including the critical case by strict convexity. At $\mu_R = 0$, $F = 1$. Thus $q = 1$ for $\mu_R \leq 1$. For $\mu_R > 1$, $F(0) > 0$, $F(1) - 1 = 0$ and $(F(z) - z)'|_{z=1} = \mu_R - 1 > 0$, so $F(z) - z < 0$ immediately to the left of one. Strict convexity gives exactly one additional root in $(0,1)$; that smaller root is q, and $1 - q > 0$. This proof relies on the single-type stationary independent-offspring model. It proves no survival criterion for a correlated or time-varying cosmogenic lineage.

A.3 Containment does not fix a likelihood direction

Proposition 3 (A finite counterexample)**.** *Let the observer-compatible histories be $O = \{a, b, c\}$ and maker-compatible histories be $M = \{b, c\} \subset O$. Assign initial-order scores $s(a) = 0$, $s(b) = 1$ and $s(c) = 2$. Every maker history then has greater order than the observer-minimal history. Let the observed event be $E = \{c\}$, and let the observer model be uniform on O. A maker model with $P(c \mid H_{\text{mak}}) = r$ and $P(b \mid H_{\text{mak}}) = 1 - r$ has*

$$\Lambda_{\mathcal{R}}(E) = 3r, \qquad 0 \leq r \leq 1. \tag{A.3}$$

The ratio is below, equal to or above one as r is below, equal to or above $1/3$. The containment and minimum-order assumptions hold throughout.

Proof. The observer likelihood of E is $1/3$ and the maker likelihood is r. Their ratio is $3r$; neither support containment nor the order scores constrain r further. □

If E were the entire set M and the maker model were exactly the observer measure conditioned on M, a different result would follow: $P(E \mid H_{\text{mak}}) = 1$. Those additional choices are absent from the generic initial-order claim. A physical likelihood needs both a specified event and a justified measure.

A.4 Normalising maker-selection

Write $Z_N = \int p_0(\theta)L(\theta)\,d\theta$ and $Z_R = \int p_0(\theta)L(\theta)M(\theta)\,d\theta$. For positive finite normalisers,

$$\frac{p(\theta \mid H_R, \text{obs})}{p(\theta \mid H_N, \text{obs})} = \frac{M(\theta)}{\mathbb{E}_{H_N,\text{obs}}[M(\theta)]} \tag{A.4}$$

on common support. Maker weighting favours a point only relative to its average weight under the comparator. For finite-resolution data the likelihood integrates this density through the observation model. No evidence for a physical selection mechanism follows from writing down the product.

B Five studies specified for registration

B.1 Study HRIH-A: model space, priors and observer measures

Question. How strongly do posterior odds depend on the model partition, prior family and xerographic distribution?

Design. Freeze all top-level families and submodels, prior families, evidence clusters, dependence structure, observer measures, prior-predictive checks and model-removal analyses before computing a posterior. Report the complete posterior surface.

Primary failure. No HRIH-favouring headline is licensed if the sign or order of the result changes under routine, predeclared alternatives.

B.2 Study HRIH-B: one physical cosmogenesis channel

Question. Does one named physical channel admit non-negligible transition probability in a regime accessible to advanced agents?

Design. Select one channel only. Freeze its action or Hamiltonian, parameters, boundary conditions, accessible-resource model, approximation validity and failure criteria. Calculate Eq. (12) or the channel-appropriate transition probability. Publish an unscored result if the required physical quantities cannot be supplied.

Primary failure. The channel is excluded, inaccessible or negligibly suppressed across the preregistered range.

B.3 Study HRIH-C: maker-selection

Question. Does maker-selection predict held-out parameter relations beyond habitability and rival selection models?

Design. Freeze L, C, S and K proxies and the held-out split before fitting. Compare predictive log density (Vehtari, Gelman, and Gabry 2017) and Bayes factors (Kass and Raftery 1995) against life-only, black-hole-production and technology-only selection. Use negative controls and leave-one-parameter-family-out sensitivity.

Primary failure. HRIH fails to exceed the smallest meaningful predictive advantage, with the interval wholly inside the equivalence region.

B.4 Study HRIH-D: cosmogenic branching

Question. Under explicit uncertainty and dependence, what posterior mass lies above $\mu_R = 1$?

Design. Build a joint probabilistic model of the factors in Eq. (15). Report posterior distributions for each factor, dependence assumptions, extinction probability and value-of-information analysis identifying

which measurements dominate uncertainty. Use Proposition 2 only for stationary independent Poisson offspring; changed offspring models require their own survival analysis.

Primary failure. For the registered stationary independent Poisson submodel, posterior mass above one is negligible across the declared joint factor distributions. A different offspring process requires its own registered failure criterion.

B.5 Study HRIH-E: closed-cycle admissibility

Question. Does a specified theory admit a stable, non-zero-measure fixed point or periodic cosmogenic history with a distinctive observable consequence?

Design. Select one explicit physical framework. State the global state space, operator C, boundary conditions, perturbation norm and measure. Prove or numerically establish existence and stability, then derive an observational discriminator.

Primary failure. No admissible solution, zero measure, instability, or no predictive distinction from the open-lineage model.

Registration status. These five study labels are retained from version 4.1 as protocol specifications. This edition supplies no new registration receipt or result. A study begins only after the author has published that unit's registration and its code and lab requirements are satisfied. A private timestamp, where documented, is described with its own date and kind; it is not an accepted public registration.

C Notation and minimum data schema

Symbol	Definition
X_g	Universe-history state in generation g
A_g	Cosmogenic agent/civilisation state
θ	Laws, constants and boundary parameters
χ	Physical cosmogenesis channel
$\mathcal{T}_\chi$	Parent-to-daughter transition kernel
C	Complete-history operator used in the closed-cycle submodel
$L(\theta)$	Observer/life permission factor
$C(\theta)$	Cosmogenic-accessibility factor
$S(\theta)$	Stable technological-ascent factor
$K(\theta)$	Viable daughter-yield factor
p_W	Choice probability for physically real cosmogenesis
μ_R	Expected number of viable daughter universes per universe history
$\xi(o)$	Xerographic or observer-weighting distribution
π_j	Prior probability of model family H_j

A minimum machine-readable record for each physical submodel should contain:

```
model_id, family, channel, state_space, dynamics_or_action,
boundary_conditions, parameter_priors, accessibility_model,
observer_measure, evidence_ids, likelihood_code_hash,
prior_predictive_status, posterior_predictive_status,
kill_condition, registration_id, version, sha256
```

D Reference implementation

The reference file `hrih_bayesian_model.py`, to be deposited with this paper, implements:

- finite-model Bayesian updating from supplied priors and relative marginal likelihoods;
- the Poisson branching extinction probability;
- the cosmogenic reproduction number in Eq. (15);
- observer-weighted created-observer probabilities and a two-model posterior sensitivity grid over priors and Bayes factors;
- basic validation and self-tests.

It contains no empirical posterior, no hidden prior and no cosmological dataset. It does not implement the maker-selection weighting of Eq. (23), which awaits the frozen C, S and K proxies of Study HRIH-C. Its purpose is to make the conditional arithmetic inspectable and to prevent prose from drifting away from equations.

E Skeleton of an analysis for registration

A registered model-space analysis should commit to the following sequence:

1. hash and freeze every dataset and code file;
2. publish model and submodel definitions;
3. publish priors and observer measures;
4. specify prior-predictive checks without running study code;
5. freeze exclusions and evidence-cluster dependence;
6. the author publishes the registration; verify its public state before running the registered checks;
7. compute marginal likelihoods;
8. run predeclared sensitivity analyses;
9. report all models, including those driven near zero;
10. publish unchanged machine output before narrative interpretation.

F Illustrative sensitivity, not evidence

Suppose three top-level models receive priors π_{N}, π_{S}, π_{R} and an evidence cluster supplies marginal likelihoods Z_{N}, Z_{S}, Z_{R}. The posterior odds for HRIH against natural cosmology are

$$\frac{P(H_{\mathrm{R}} \mid D)}{P(H_{\mathrm{N}} \mid D)} = \frac{\pi_{\mathrm{R}}}{\pi_{\mathrm{N}}} \frac{Z_{\mathrm{R}}}{Z_{\mathrm{N}}}. \tag{F.1}$$

This identity shows why no amount of prose can supply a posterior. Both prior odds and a likelihood ratio are required. If the evidence is equally expected under the models, $Z_{\mathrm{R}}/Z_{\mathrm{N}} = 1$ and the posterior odds remain the prior odds. For broad ARC correspondences the present convention is no update, as Section 9 explains; equal empirical likelihoods have not been established.

G Programme placement and the five laws under test

G.1 Separate scientific objects

The ARC Theory statement paper, version 7.6, treats recursive capability, correction and value persistence; the Recursive Dynamics founding paper sets out the field. HRIH concerns cosmological origin and reproduction. Neither requires the other's truth. If HRIH fails, the ARC laws retain their own registered tests. If an ARC law fails, HRIH loses only a proposed stability model that actually uses it. Section 9 explains why broad ARC data currently make no cosmological update.

ARC-derived work could inform $S(\theta)$ or p_K, the proposed probability that technological intelligence remains stable enough to complete cosmogenesis. Such a bridge would require a model; it would not establish that our universe was selected. The Eden Protocol and the HARI Treaty, the Draft Treaty on Hardware-Aligned Recursive Intelligence, provide engineering and governance strands of the wider programme; neither is a cosmogenic likelihood in this paper. Humans retain control as long as possible. Embedded correction is a contingency for involuntary loss of control, not a plan to surrender it.

G.2 The five laws and their distinct status

The following synopsis uses the statement paper at version 7.6 and the current theory registration, version 1.104. Laws I to III were registered on 8 September 2026; Laws IV and V were added on 24 September and registered as laws on 15 October in version 1.103. All five remain laws under test. Their scientific roles, rather than obsolete verbatim formulations, are retained here:

Law I, the ARC Principle. $U = IR^{\alpha}$ is a heuristic capability form with a measured exponent. Precursor error-decay exponents are not capability exponents or a registered verdict on this law.

Law II, the ARC Co-Scaling Law. $\beta > k$ gives relative drift an asymptotic disadvantage in the positive-gain minimal model. A decreasing drift fraction does not by itself bound absolute or cumulative drift.

Law III, the ARC Ceiling. $\alpha_{\text{crit}} = 1/(1-\gamma)$ is a conditional pacing-model relation. Real-system applicability and the additional premise giving the ARC Bound of two remain under test; no universal ceiling follows from the formula alone.

Law IV, the ARC Persistence Law. P15 concerns retained externally installed gain, $S_{\text{ext}}(1) < 1/2$, after matched capability-training budget. Version 1.104 also states a comparison limb; that limb has no registered proposition yet and is not evaluable. The law is untested; its mathematics, version 1.2, remains in development.

Law V, the ARC Embedding Law. The build-order limb requires the directed difference $\gamma_{\text{first}} - \gamma_{\text{after}} \geq 0.10$; placement asks whether the inside-loop advantage over output correction widens with depth. Unit 83 gives a qualified, family-dependent result on the access axis of P21, with its mechanism axis not evaluable. It neither isolates information access from repair rights nor supports Law V as a whole. The Unit 83 results carry the outcome; the mathematics remains in development.

None of these laws enters a cosmological likelihood here. P12, P15 and P21 are propositions of the theory registration; this paper's P1 to P5 are separate labels.

G.3 The conditional ARC Bound

Within $\alpha_{\text{crit}} = 1/(1-\gamma)$ and the domain $\gamma < 1$, the bound $\alpha_{\text{crit}} \leq 2$ is equivalent to $\gamma \leq 1/2$. This algebra supplies no evidence for either empirical premise. The domain is essential: outside it the same monotonic comparison cannot be used. HRIH remains neutral on the relation's real-system applicability.

G.4 The long-form cultural and ethical material

The existing long-form HRIH edition contains the genesis narrative, religious convergence, ethical argument, future ladder, priority record and Eden governance. Those materials are preserved as dated work. They are not used as cosmological evidence here. In a publication architecture they should be cited through separate provenance, reception and ethics companions so that the scientific paper neither deletes them nor asks them to carry a likelihood they do not possess.

H Simulation and real cosmogenesis: comparison table

H.1 Different explanatory targets

The simulation hypothesis proposes that the experienced world is implemented on an external computational substrate. HRIH-R1 proposes that a physically real daughter cosmology is instantiated. HRIH-R3 further proposes a global closure of the cosmogenic history. Simulation can explain why an apparent world has chosen features if a simulator selected them, but ordinarily transfers the origin question to the simulator's level. Closed-cycle HRIH attempts to close that regress, at the cost of a much more demanding physical and measure-theoretic burden.

Section 4.4 states the conditional observer-count identity; Section 6.6 distinguishes the resource models. Neither supplies measured odds for the broad hypothesis families.

H.2 Physical constraint comparison

Table 5 compares the two families dimension by dimension.

Dimension	Simulation family	Real intelligent cosmogenesis
Substrate	External computing system	Parent physical universe and cosmogenic channel
Resource question	Cost and fidelity of implementing observed experience	Cost, action and accessibility of seeding a daughter state
Observer problem	Which simulated observers are in the reference class	Which observers and universe generations receive weight
Distinctive burden	Number and fidelity of simulations	Existence, control and viability of a real channel
Origin regress	Usually transferred to base reality	Open lineage transfers it; closed-cycle submodel attempts closure
Present status	Philosophically developed, physically disputed	Physically and statistically underdeveloped

Table 5. The hypotheses require different likelihoods. Neither earns probability from public familiarity or the number of articles discussing it.

H.3 Present comparison

This paper supplies no robust numerical comparison between the broad families. Specific simulation or cosmogenesis submodels can face physical constraints; the lack of a scored HRIH model is not a demonstration that all public evidence fails to discriminate every such submodel. The required comparison is between specified likelihoods and observer measures, not unbounded family names.

I Eight rungs of claim, and the neighbour each proposition degrades to

Rung	Claim	Present status	What moves it
1	Advanced agents may model or simulate cosmologies	Established in weak computational forms; not universe creation	Better simulation and theoretical work
2	A physically admissible daughter-universe channel exists	Open; serious theoretical precedents, major obstacles	HRIH-B and mature quantum gravity
3	Agents can control such a channel	Open and more demanding than existence	Parent-to-daughter parameter map
4	Agents choose real cosmogenesis	Open normative/behavioural parameter	Evidence bearing on p_W
5	Viable daughter universes result	Open	Channel physics and viability model
6	Multi-generation intelligent lineages occur	Open; conditional branching model supplied	Posterior on μ_R
7	Maker-selection shaped our constants	Unconfirmed; principal empirical target	HRIH-C held-out prediction
8	The lineage forms a globally closed cycle	Most speculative; postulated, not derived	HRIH-E existence, stability and prediction

I.1 What remains when a commitment fails

The following dispositions complement K1 to K7. They distinguish rejection of the current claim from a separately named, weaker scientific object:

Computation replaces instantiation. The result belongs to the simulation family; Bostrom's argument is a prominent form of that family, not a proof that our universe is computed (Bostrom 2003).

Reproduction has no agent. The result belongs to natural cosmological reproduction; Smolin's black-hole selection is one specific predecessor (Smolin 1992).

Global closure fails. An open lineage may remain, but the claimed closed-cycle explanation is rejected.

Multiple generations fail. One-generation intelligent cosmogenesis may remain, adjacent to the creation and selection proposals of Crane, Harrison and Gardner (Crane 1994; Harrison 1995; Gardner 2003). No threshold or lineage-survival result is inferred for that single event.

Maker-selection fails. A channel-feasibility question may remain; the distinctive maker-selection claim is rejected, not renamed a success.

The first two changes remove the defining physical-instantiation or agent conditions. The others remove the specified stronger commitment. Any successor hypothesis receives its own date and statement, as Section 11 requires.

J Version history and corrections

J.1 Separation of provenance from evidence

The earlier HRIH materials contain a priority and genesis record, which is authorship and historical evidence rather than cosmological evidence. The deposited manuscript bundle (Eastwood 2024) supports the dated manuscript claim only. No deposited priority-index DOI is asserted here, and priority for the exact conjunction remains unresolved.

J.2 Distinct editions

The long-form HRIH monograph and this Bayesian research programme have separate identities and DOI records. This edition does not replace the monograph. The comparison below is explicitly against its version 3.51 as read for version 4.1; it is a historical comparison, not a claim about every later monograph edition.

J.3 Version 4.2, 2 November 2026

The Paper X template replaces the designed cover and boxes. The abstract and introduction begin on page 1; contributions and related work move to the opening sections; proofs, detailed study specifications, placement and history are lettered appendices. Repeated prior, status and compatibility passages are consolidated. Substantive corrections distinguish conditional observer counting from a posterior ceiling, make cost derivatives model-specific, withdraw the asserted positive entropy likelihood ratio, state the Poisson assumptions, normalise maker-selection and mark ARC neutrality as a no-update convention. The five-law synopsis uses the current statement and registration. All earlier released bytes remain unchanged.

J.4 Version 4.1, 21 October 2026

Corrected the law-registration status and Law II boundary wording against registration 1.103, and cited the then-current mathematics of Laws IV and V. The first website publication of this edition was 22 October 2026. The earlier reconstruction and its reviews remain dated records.

J.5 Material relocated, not erased

The bodies of work taken out of the evidential argument remain valuable but carry different evidential classes. Appendix L lists each with its treatment here and its continuing home.

J.6 Corrections carried from the long-form record, which are the ones that cost something

The following three entries preserve the correction record carried by version 4.1. They document revisions, not evidence for the cosmological hypothesis; current precursor interpretation is stated with the first entry.

1. The exponent originally reported at approximately 2.24 was the programme's flagship number. It is retracted. Paper II (Eastwood 2026f) reports, on Gemini 3 Flash, the model it treats as its most reliable estimate, an endpoint sequential error-decay exponent of 0.59 with a bootstrap 95 per cent interval of −1.3 to 2.9, a regression sequential error-decay exponent of 0.49 with a standard error of 0.20 and a coefficient of determination of 0.86, and a compute-matched parallel error-decay exponent of 0.31. These are one fixed model's precursor fits on the error-decay axis, with a wide interval. They do not estimate the capability exponent or test the ARC Ceiling; the numbers and their uncertainty are retained as the historical correction record.

2. A four-term decomposition comparing this hypothesis with the simulation hypothesis was withdrawn in place. It was the only quantitative comparison the programme could make with its best-known rival, and withdrawing it left no numerical claim against that rival in either direction.
3. In the twelve-entry record reported by version 4.1, eleven entries were withdrawn pending audit and the twelfth was retired on 6 September 2026 because its cited result predated it. That historical statement concerns those twelve entries, not the programme's present registrations or later prospective predictions.

J.7 Corrections made by this reconstruction

This reconstruction records, rather than conceals, the following changes in scientific position:

1. recursion no longer supplies a proof against a first boundary;
2. occurrence is separated from constructibility;
3. creator ability is separated from preference;
4. lower-level ARC results are assigned neutral cosmological evidence absent a bridge;
5. numerical HRIH probabilities are withheld pending registered models;
6. compatibility claims are not counted as support;
7. the closed loop is a nested, stronger submodel rather than the premise of all intelligent cosmogenesis.

J.8 Where this paper and the long-form edition differ

The list below records what version 4.1 changed relative to the long-form edition at version 3.51. It remains useful provenance. The v4.2 qualification at the end governs the present treatment of the observer count, duration derivative and entropy ratio.

1. **The limit of the observer count.** Its section 12A.3 takes the complement family to make no created observers; under this paper's three families two families create, so the limit is stated for each creating family, and under equal weights the chance of having been made by either tends to two thirds (Section 4.4).
2. **Vazza's figures.** Its section 12A.4 gives 5.6×10^{59} erg as the cost of initialising a simulation of the Earth and has the bill reach the Milky Way's rest-mass energy after a handful of steps; his own figures are 2.55×10^{59} erg, for the Earth at full resolution, and about 10^6 timesteps, and 5.6×10^{59} erg is his comparison figure for Palomar 2.
3. **The source of the derivative.** Its section 12A.4 credits the whole duration derivative to Vazza's per-timestep result; only the derivative's first half is his, and its second half is this paper's premise (Section 6.6).
4. **The resource standing of instantiation.** Its section 12A.4 calls this paper's mechanism unobstructed on resources; this paper claims only that no bill grows with the child's duration, and its one-off cost is not yet calculated.
5. **The two obstacles.** Its section 12A.4 places a hypothesis blocked by an open question in a better epistemic position than one blocked by a closed sum; this paper claims a difference in kind and no better position.
6. **The reach of the derivative.** Its section 12A.4 says that an impossibility-class result against simulation says nothing whatever about instantiation; here the derivative answers resource-budget results only, and the undecidability result, which the long-form edition also answers on its own ground, is answered separately.

7. **The reproduction number.** Its section 12A.10 defines m_R as the expected viable daughters given success, which beside p_V counts viability twice; here m_R counts the daughters generated (Eq. (15)), as the reference code does.
8. **The argument from the initial order.** Its section 12A.5 calls the sign of the ratio a consequence of the containment relation that is not model-dependent, has raising require a global thermodynamic gradient sustained across cosmic time, and has its K2 collapse the containment relation; here the sign rests also on the premise that raising needs more initial order than observing, which E2 would defeat, and raising needs a gradient only for as long as raising takes (Section 8.5).
9. **Penrose's resolution.** Its section 12A.5 names conformal cyclic cosmology as Penrose's resolution and calls maker-selection the second hypothesis his argument has lacked; in the 1979 chapter from which this paper quotes his argument, his resolution, and so his own second hypothesis, is the initial vanishing of the Weyl curvature, which conformal cyclic cosmology later carried, and this paper names both and calls maker-selection a second selection hypothesis.
10. **The laws' place in the hypothesis.** Its section 1D says that if Law I fails "the capability premise beneath the whole construction goes, and HRIH goes with it", and calls the dependence on Law II "the most load-bearing single dependency"; here none of the five laws is a premise of HRIH, and a law's failure takes away at most the creator-stability model it informs (Section G), as the statement paper, at version 7.3, holds of HRIH: "no outcome of any register is evidence for or against it".
11. **The two definitional concessions.** Its section 15.1, on which of its seven claims can be lost, gives the computation case to Bostrom and the no-mind case to Smolin; Bostrom argues a trilemma rather than that our universe is a computation, and Smolin proposes one mechanism with no mind involved, so this paper gives the two cases to the simulation family and to natural cosmological reproduction and cites Bostrom's argument and Smolin's selection through black holes as the best-known form of each (Section I.1).

The compared long-form edition also counted all twelve tracked entries as withdrawn; Section J.6 preserves the later historical classification.

Version 4.2 qualification. The old list's observer-count limit holds only for the explicit mixture in Proposition 1; its duration derivative holds only under the cost assumptions of Section 6.6; and its claim to a favourable entropy likelihood direction is withdrawn by Section 8.5. Those earlier descriptions are retained here as dated correction history, not adopted current results.

K Priority record

The hypothesis is named in the author's self-emailed manuscript record of 8 December 2024. That is dated provenance, not evidence for cosmogenesis or a contemporaneous public publication. The manuscript bundle has been public on OSF since 5 October 2026. The long-form edition was first published on 3 July 2026; this Bayesian companion has its own DOI, doi:10.17605/OSF.IO/4PGVC.

The source email hash reported by version 4.1 is f0d1f38ffd8546152d9d9d28dc5ec083c16a35858f2c12b63e69db7ed50901ad. A hash identifies bytes; its dating force comes from the separately retained message and deposit records. This edition does not claim a new timestamp verification. Earlier universe-creation, observer-selection and reproductive-cosmology work is credited in Section 2; priority for the six-part conjunction remains open review.

L Version transition from the long-form edition

Long-form material	Scientific-reconstruction treatment	Continuing home
Closed causal loop	Retained as nested R3; fixed-point formalism; no derivation from recursion	This paper
Intelligent cosmogenesis	Retained as R1 with channel-specific feasibility	This paper
Recursive universe lineage	Retained as R2 branching model	This paper
Fine-tuning narrative	Replaced by a maker-selection likelihood to be registered	This paper
Priority dossier	Removed from evidential argument, preserved and cited	Provenance register
Religious convergence	Preserved, labelled interpretive/non-evidential	Companion essay
Personal genesis narrative	Preserved as first-person provenance	Book/essay
Eden governance	Preserved and separated from cosmological evidence	ARC/Eden ethics paper
Future ladder	Preserved, labelled forecast/non-evidential	Forecast register
Cross-domain compatibility	Retained only as compatibility, never support	Companion/reference pages
Numerical HRIH probability	Withdrawn unless generated by registered model	HRIH-A future result

Where the third column names a home that does not yet exist, the material's home today is the long-form edition, where it is preserved as dated work.